Three initial ideas:

1. Game recommendation system

What it does: Recommends a game based on user responses to questions like genre, difficulty, or your current mood.

How it works using rules: Use if/else rules. For example, if the user selects competitive and difficult, the program recommends Apex Legends. If the user selects relaxing and single player, the program recommends Cyberpunk

2. Tech troubleshooting assistant

What it does: Solves a tech issue by asking yes or no questions to suggest next steps and eventually solve the issue.

The system follows predefined cases. For example, if your computer doesn't turn on suggest checking the power cable. If your monitor doesn't display an image, suggest inspecting the display or power cable.

3. Food recommendation system

What it does: Recommends a meal based on diet preference, hunger level, and time available to cook

This system uses decision rules. If a user selects high protein and has less than an hour to cook, suggest chicken and rice.

My choice: Troubleshooting Assistant

I chose this rule-based system because I recently went through a subtle technical issue with my gaming PC. If I had a system that could identify symptoms and provide accurate recommendations to solve them, I would have saved myself money and a year of stress. Now that I have gone through the troubleshooting process, I have a clear idea of what I would need for my system.

Rules/Logic:

If user inputs **"won't turn on", "no power", "dead"**

→ Ask if the power cable is plugged in and suggest checking the outlet, PSU switch, and power button connection.

If the user says **"turns on but no display", "black screen", or "no signal"**

→ Suggest checking the monitor cable, making sure the monitor is on the correct input, and verifying the GPU/display cable connection.

If the user says **"slow", "laggy", or "freezing"**

→ Suggest closing background programs, restarting the computer, checking Task Manager, and scanning for updates or malware.

If the user says **"internet", "wifi", or "connection"**

→ Suggest restarting the router, checking Wi-Fi connection, testing Ethernet, or running Windows network troubleshooting.

If the user says **"overheating", "hot", or "shutting down"**

→ Suggest checking fan operation, cleaning dust, improving airflow, or checking CPU/GPU temperatures.

If the user says **"blue screen", "crash", or "restart"**

→ Suggest noting the error code, checking recent driver updates, removing newly installed software, or running system diagnostics.

Testing Inputs:

Mouse input lag related problems:

What problem are you having? mouse laggy

Recommendation:

Close background programs, restart the computer, check Task Manager, and scan for updates or malware.

Thank you for using the Troubleshooting Assistant!

No sound or audio:

Welcome to the Troubleshooting Assistant!

Describe your computer issue and I will suggest a possible solution.

Example: no power, no display, slow computer, internet issue, overheating, or blue screen

What problem are you having? no audio

Recommendation:

Try disconnecting and reconnecting your device, checking sound settings, verifying cables are plugged in, and ensuring the correct playback device is selected

Thank you for using the Troubleshooting Assistant!

Power-related problems:

Welcome to the Troubleshooting Assistant!

Describe your computer issue and I will suggest a possible solution.

Example: no power, no display, slow computer, internet issue, overheating, or blue screen

What problem are you having? no power

Recommendation:

Check the power cable, wall outlet, power supply switch, and power button connection.

Thank you for using the Troubleshooting Assistant!

Reflection:

My rule-based system works as a guide to basic technical issues a user might be having. It works by identifying keywords that the user inputs and offering advice based on those inputs. My program prompts the user to describe their issue briefly and shares an example to hopefully convey the expected simplicity of the user's response. Then, the user will type some input based on what issues they are having. Within their response if one of the many keywords are triggered, it will respond with recommendations to solve their technical issue. If the user's input does not contain any keywords in the program it will prompt the user to describe their issue with different keywords and prompt them with the initial question again.

Overall, I didn't encounter too many issues developing this program with AI. I was able to generate plenty of valuable ideas that I already relate to because of my prompts to this agent in the past. One challenge I encountered was making sure the code generated matches the assignment's requirements. The agent was able to generate a working rule-based system quickly, but I still needed to review the code and add additional rules that offered a broader umbrella to the many technical issues possible.

This activity helped me understand how rule-based systems depend on the rules that were programmed, so if an issue does not match any of the predefined keywords or rules, the system cannot offer a recommendation. This is helpful because it shows why modern AI is so beneficial. It uses advanced techniques like machine learning to learn from data and make predictions without being restricted by fixed rules that are programmed by a developer.